(i)

$$\int_0^\infty e^{-st}e^{-bt}f(t)\,dt = \int_0^\infty e^{-(s+b)t}f(t)\,dt = F(s+b)$$

(ii) Let $u = at,\ du = adt$

$$\int_0^\infty e^{-st}f(at)\,dt = \frac{1}{a}\int_0^\infty e^{-(s/a)u}f(u)\,du = a^{-1}F\big(\frac{s}{a}\big)$$

(iii)

$$\int_0^\infty e^{-st}f'(t)\,dt = \big[e^{-st}f(t)\big]_0^\infty + s\int_0^\infty e^{-st}f(t)\,dt = sF(s) - f(0)$$

(iv)

$$\begin{aligned}
F(s) &= \int_0^\infty e^{-st}\sin t\,dt \\
&= \big[-e^{-st}\cos t\big]_0^\infty - s\int_0^\infty e^{-st}\cos t\,dt \\
&= 1 - s\int_0^\infty e^{-st}\cos t\,dt \\
&= 1 - s\left(\big[e^{-st}\sin t\big]_0^\infty + s\int_0^\infty e^{-st}\sin t\,dt\right) \\
&= 1 - s^2F(s)
\end{aligned}$$

$$\implies \boxed{F(s) = \frac{1}{s^2+1}}$$

Let $f(t) = \sin t,\ f'(t) = \cos t,\ f(0) = 0$

$$\int_0^\infty e^{-st}f(t)\,dt = \frac{1}{s^2+1}$$

$$\int_0^\infty e^{-st}f'(t)\,dt = s\cdot\frac{1}{s^2+1} - 0 = \frac{s}{s^2+1}$$

$$\int_0^\infty e^{-st}f'(qt)\,dtx = \frac{1}{q}\cdot\frac{\frac{s}{q}}{\big(\frac{s}{q}\big)^2+1} = \frac{s}{s^2+q^2}$$

$$\int_0^\infty e^{-st}e^{-pt}\cos(qt)\,dt = \frac{s+p}{(s+p)^2+q^2}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

Last updated: June 25, 2026